

Algorithms Problem Solving

Level 1

Grade A B C D E F

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Definition

- ◆ This program asks the user to enter a grade between 0 and 100 and converts the numeric grade into a letter grade.
- ◆ The possible results are A, B, C, D, E, and F.
- ◆ It uses a `do-while` loop to make sure the entered grade is within the valid range.

Explanation

- ◆ ``readGrade()`` asks the user for a grade between ``From`` and ``To``.
- ◆ The ``do-while`` loop repeats the input while the grade is outside the valid range.
- ◆ ``grade < From || grade > To`` means the input is invalid if it is below the minimum OR above the maximum.
- ◆ ``getGrade()`` converts the numeric grade into a character.
- ◆ The ``if-else if`` statements check the ranges from highest to lowest.
- ◆ Because the conditions use ``>=``, each boundary belongs to the higher grade.
- ◆ ``printResult()`` calls ``getGrade()`` and prints the returned character.
- ◆ ``printResult(readGrade(0, 100))`` first reads a valid grade and then passes it to ``printResult()``.

Steps in main()

1. Ask the user to enter a grade between 0 and 100.
2. Repeat the input if the grade is outside the valid range.
3. Check if the grade is 90 or higher.
4. If not, check 80, then 70, then 60, then 50.
5. Return the corresponding letter grade.
6. Return `F` if the grade is below 50.
7. Print the final letter grade.

💡 Example

◆ If the user enters:

Grade: 85

◆ The program checks:

$85 \geq 90 \rightarrow \text{false}$

$85 \geq 80 \rightarrow \text{true}$

➡ Result

Result = B

Note

◆ The grade ranges are:

A → 90-100

B → 80-89

C → 70-79

D → 60-69

E → 50-59

F → 0-49

◆ The conditions only need to check the lower boundary because the previous condition has already been checked.

⊘ Warning

◆ The program validates that the grade is between 0 and 100, but it does not handle non-numeric input such as letters.



Comparison

Valid Grade

- ✓ Between 0 and 100.
- ✓ Accepted by ``readGrade()``.
- ✓ Converted into a letter grade.

Invalid Grade

- ✓ Less than 0, or
- ✓ Greater than 100.
- ✓ The program asks the user to enter the grade again.

✦✦ Summary

- ✓ The program converts a numeric grade into A-F.
- ✓ `do-while` validates that the grade is between 0 and 100.
- ✓ `if-else if` checks the grade ranges from highest to lowest.
- ✓ `>=` makes the lower boundary of each grade inclusive.
- ✓ The function returns a `char` such as `A`, `B`, `C`, `D`, `E`, or `F`.

Thank you 😊